

Synthesizing Grid Data with Cyber Resilience and Privacy Guarantees

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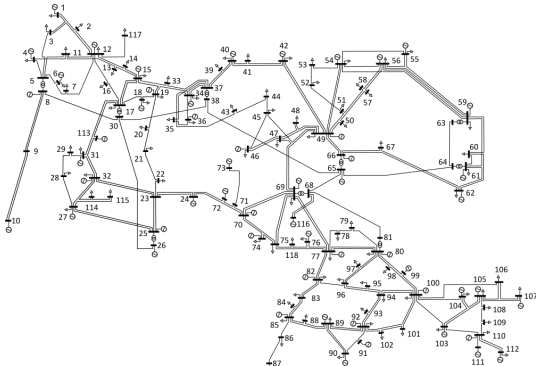
- ▶ Power systems are critical infrastructures with most of data being classified
- ▶ We have only a limited observability, e.g., ISO data disclosure portals
- ▶ Grid stakeholders hence act on a limited set of system data
- ▶ Research lags behind due to lack of real system data



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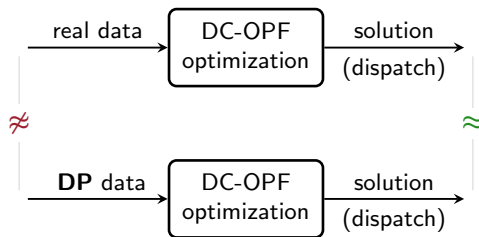


Example: DC optimal power flow (OPF) in the (small) IEEE 118-bus system



- ▶ 1079 rows of element-specific data
- ▶ Each generator owns only 2 rows
- ▶ The rest of system data is not *explicitly* disclosed to power producers

**How to disclose grid data
in a controllable manner?**



- ▶ DP principle: obfuscate real data (add noise) but preserve its quality
- ▶ In the DC-OPF setting: obfuscate grid data but preserve the OPF solution
- ▶ Formal *privacy guarantee*: released DP data does NOT disclose the real data
- ▶ Many applications to synthesizing high-quality transmission, load and generation data

DC-OPF Formulation:

$$\begin{aligned}
 C_{\text{opf}}(\mathbf{d}) = \min_{\mathbf{p}, \mathbf{v} \geq 0} \quad & \mathbf{q}^\top \mathbf{p} + \psi^\top \mathbf{v} && \text{overall generation cost} \\
 \text{s.t.} \quad & \underline{\mathbf{p}} \leq \mathbf{p} \leq \bar{\mathbf{p}} && \text{generation capacity} \\
 & \mathbf{1}^\top (\mathbf{p} - \mathbf{d}) = 0 && \text{power balance} \\
 & |\mathbf{F}(\mathbf{p} - \mathbf{d})| \leq \bar{\mathbf{f}} + \mathbf{v} && \text{line capacity}
 \end{aligned}$$

Compact Form:

$$\begin{aligned}
 C_{\text{opf}}(\mathbf{d}) = \min_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} \quad & \mathbf{a}_k^\top \mathbf{x} + \mathbf{b}_k^\top \mathbf{d} + e_k \leq 0, \\
 & \forall k = 1, \dots, K,
 \end{aligned}$$

Optimization Parameters

- $\mathbf{q}$: generator cost
- $\mathbf{d}$: load data
- ψ : penalty parameter
- $\mathbf{F}$: PTDF Matrix
- $\bar{\mathbf{p}}, \underline{\mathbf{p}}$: generation limits
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How to disclose a DP version of load data?

DC-OPF:

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How to disclose a DP version of load data?

- ▶ Step 1 (privacy guarantee):

$$\begin{aligned} \tilde{\mathbf{d}}^0 &= \mathbf{d} + \text{calibrated noise}, \\ \tilde{C}_{\text{opf}} &= C_{\text{opf}}(\mathbf{d}) + \text{calibrated noise} \end{aligned}$$

- ▶ Step 2 (post-processing for fidelity):

$$\min_{\tilde{\mathbf{d}}} \|C_{\text{opf}}(\tilde{\mathbf{d}}) - \tilde{C}_{\text{opf}}\|_1 + \|\tilde{\mathbf{d}} - \tilde{\mathbf{d}}^0\|_1$$

- ▶ Grid datasets are used for calibrating cyberattacks on power grids
- ▶ Hypothesis: high-quality synthetic data → well-calibrated attacks
- ▶ Classes of cyberattacks: false data injection, line outage masking, physical attacks, ...

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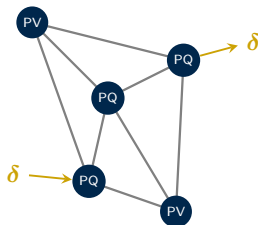
Contribution: We identify cyberattack risks in releasing DP grid data and propose new algorithms to guarantee both privacy and cyber resilience to source grids

- Given load $\mathbf{d}$, the attack corrupts the load $\mathbf{d} + \boldsymbol{\delta}$ with bus injection $\boldsymbol{\delta}$ from admissible set Δ

$$\Delta \triangleq \left\{ \boldsymbol{\delta} \mid \begin{array}{l} \underline{\boldsymbol{\delta}} \leq \boldsymbol{\delta} \leq \bar{\boldsymbol{\delta}} \\ \mathbf{1}^\top \boldsymbol{\delta} = 0 \end{array} \quad \begin{array}{l} \text{injection limits for each bus} \\ \text{total load remains unchanged} \end{array} \right\}$$

- Amounts solving a bi-level optimization problem:

$$\begin{aligned} C_{\text{att}}^{\text{BO}}(\mathbf{d}) &= \max_{\boldsymbol{\delta} \in \Delta} C_{\text{opf}}(\mathbf{d} + \boldsymbol{\delta}) && \text{maximize the cost} \\ \text{s.t. } C_{\text{opf}}(\mathbf{d} + \boldsymbol{\delta}) &= \min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x} && \text{feedback from OPF} \\ \text{s.t. } \mathbf{a}_k^\top \mathbf{x} + \mathbf{b}_k^\top (\mathbf{d} + \boldsymbol{\delta}) + e_k &\leq 0 \end{aligned}$$



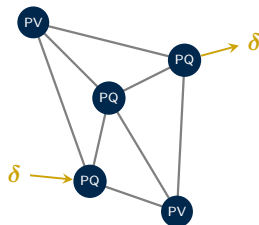
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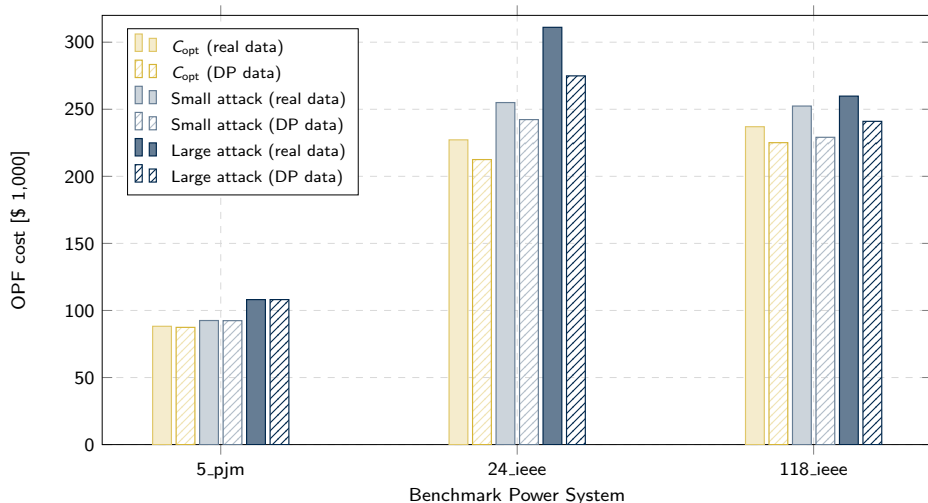
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Can DP grid data be used to successfully execute the load redistribution attack?

Average Outcomes of Load Redistribution Attacks



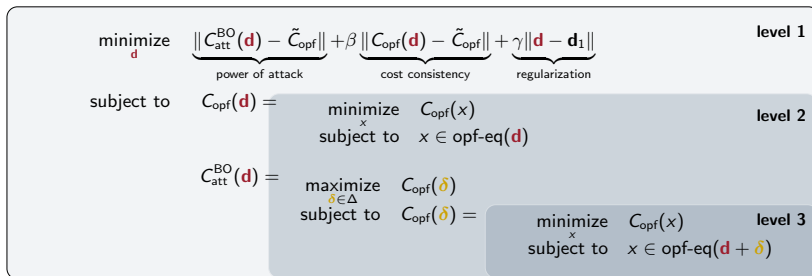


Figure: Tri-level structure of the cyber-resilient post-processing of DP load data.

- **Step 1:** add random noise to real loads: $\mathbf{d}_1 = \mathbf{d} + \text{calibrated noise}$
- **Step 2:** post-process $\mathbf{d}_1$ by solving a tri-level optimization:
 - Level-1: Optimize synthetic load $\mathbf{d}$ to balance attack damage and cost-consistency
 - Level-2: Feedback from both OPF and attack optimization
 - Level-3: Embedded OPF for attack calibration
- **Result:** DP load vector $\mathbf{d}$ balancing attack damage and cost-consistency

- ▶ Optimizing synthetic loads over bi-level optimization is computationally challenging
- ▶ We drawn on the connection between bi-level and robust (single-level) optimization

Bi-level attack optimization

$$\begin{aligned} C_{\text{att}}^{\text{BO}}(\mathbf{d}) &= \max_{\boldsymbol{\delta} \in \Delta} C_{\text{opf}}(\mathbf{d} + \boldsymbol{\delta}) \\ &\quad \mathbf{x} \in \underset{\mathbf{x}}{\operatorname{argmin}} \mathbf{c}^{\top} \mathbf{x} \\ &\quad \text{s.t. } \mathbf{a}_k^{\top} \mathbf{x} + \mathbf{b}_k^{\top} (\mathbf{d} + \boldsymbol{\delta}) + e_k \leq 0 \quad \forall k \\ &\quad \text{(uniform attack)} \end{aligned}$$

Robust optimization (RO) approximation

$$\begin{aligned} C_{\text{att}}^{\text{RO}}(\mathbf{d}) &= \min_{\mathbf{x}} \mathbf{c}^{\top} \mathbf{x} \\ &\quad \text{s.t. } \max_{\boldsymbol{\delta}_k \in \Delta} \left[\mathbf{a}_k^{\top} \mathbf{x} + \mathbf{b}_k^{\top} (\mathbf{d} + \boldsymbol{\delta}_k) + e_k \right] \leq 0 \quad \forall k \\ &\quad \text{(per constraint attack)} \end{aligned}$$

Proposition: For any feasible load $\mathbf{d}$, relation $C_{\text{att}}^{\text{RO}}(\mathbf{d}) \geq C_{\text{att}}^{\text{BO}}(\mathbf{d})$ holds.

- **Step 1:** Obfuscate real load data

DP load $\mathbf{d}_1 = \mathbf{d} + \text{calibrated noise}$

DP estimation of OPF costs: $\tilde{C}_{\text{opf}} = C_{\text{opf}}(\mathbf{d}) + \text{calibrated noise}$

- **Step 2:** Post-process $\mathbf{d}_1$ to balance cost-consistency and cyber resilience P

$$\underset{\mathbf{d}}{\text{minimize}} \quad \underbrace{\|C_{\text{att}}^{\text{RO}}(\mathbf{d}) - \tilde{C}_{\text{opf}}\|}_{\text{power of attack}} + \beta \underbrace{\|C_{\text{opf}}(\mathbf{d}) - \tilde{C}_{\text{opf}}\|}_{\text{cost consistency}} + \gamma \underbrace{\|\mathbf{d} - \mathbf{d}_1\|}_{\text{regularization}}$$

- Replacing the two embedded optimization problems with their Karush–Kuhn–Tucker conditions leads to a single-level mixed-integer problem.

$$\begin{aligned}
 C_{\text{att},\tau}^{\text{RO}}(\mathbf{d}) = \min_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} \\
 \text{s.t.} \quad & \max_{\delta_k \in \Delta} \left[\mathbf{a}_k^\top \mathbf{x} + \mathbf{b}_k^\top (\mathbf{d} + \delta_k) + e_k \right] \leq 0 \quad \forall k \in \mathcal{K}' && \text{RO-reformulated cons} \\
 & \left[\mathbf{a}_k^\top \mathbf{x} + \mathbf{b}_k^\top \mathbf{d} + e_k \right] \leq 0 \quad \forall k \in \mathcal{K} && \text{original problem cons}
 \end{aligned}$$

- ▶ We select only most important τ constraints for RO reformulation
- ▶ Most important constraints in $\mathcal{K}'$ are function of original loads
- ▶ Exponential mechanism of DP to obfuscate loads when selecting $\mathcal{K}'$

$$C_{\text{att},\tau}^{\text{RO}}(\mathbf{d}) = \min_{\mathbf{x}} \mathbf{c}^{\top} \mathbf{x}$$

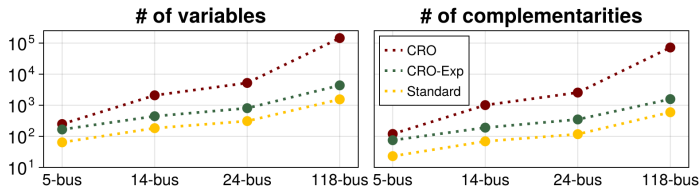
$$\text{s.t. } \max_{\delta_k \in \Delta} \left[\mathbf{a}_k^{\top} \mathbf{x} + \mathbf{b}_k^{\top} (\mathbf{d} + \delta_k) + e_k \right] \leq 0 \quad \forall k \in \mathcal{K}'$$

RO-reformulated cons

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Selecting only important constraints substantially reduces the computational burden

Algorithm 1: Privacy-preserving CRO-Exp

Input: real load $\mathbf{d}$, DP parameters $(\alpha, \varepsilon_1, \varepsilon_2, \varepsilon_3)$, attack data $(\beta, \gamma, \Delta, \tau)$, $\mathcal{K} = \{\emptyset\}$
 1 DP obfuscation of load and OPF costs:

$$\tilde{\mathbf{d}}^0 = \mathbf{d} + \text{Lap}\left(\frac{\alpha}{\varepsilon_1}\right)^n$$

$$\tilde{C}_{\text{opf}} = C_{\text{opf}}(\mathbf{d}) + \text{Lap}\left(\frac{\alpha\bar{C}}{\varepsilon_2}\right)$$

2 DP estimation of the set $\mathcal{K}$ of the worst-case constraints

for $t = 1, \dots, \tau$ **do**

for $k = 1, \dots, K$ **do**

$$C_k = C_{\text{att},t}^{\text{RO}}(\mathbf{d}) + \text{Lap}\left(\frac{\alpha\bar{C}}{\varepsilon_3}\right)$$

end

$$k_t \leftarrow \arg\max_k C_k$$

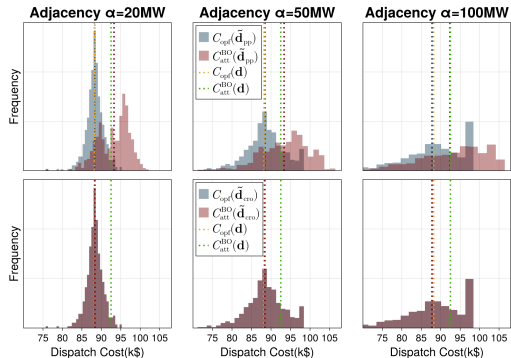
$$\mathcal{K} \leftarrow \mathcal{K} \cup \{k_t\}$$

end

3 Post-processing optimization of the synthetic load vector

$$\tilde{\mathbf{d}} \in \arg\min_{\tilde{\mathbf{d}}} \|C_{\text{opf}}(\tilde{\mathbf{d}}) - \tilde{C}_{\text{opf}}\|_1 + \beta \|C_{\text{att},\tau}^{\text{RO}}(\tilde{\mathbf{d}}) - \tilde{C}_{\text{opf}}\|_1 + \gamma \|\tilde{\mathbf{d}} - \tilde{\mathbf{d}}^0\|_1$$

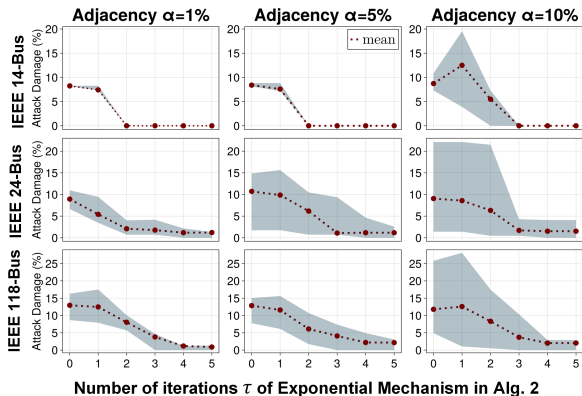
Output: Synthetic load vector $\tilde{\mathbf{d}}$



- ▶ Blue distributions - normal operation
- ▶ Red distributions - post-attack operation
- ▶ Top row - standard DP post-processing
- ▶ Bottom row - proposed CRO post-processing

CRO sends important signal to attackers: **the attacks do not lead to extra OPF cost to the system.**
(normal and post-attack distributions overlap)

OPF cost distributions in normal and post-attack operation



- The more constraints under attack → more resilient the source grid to load redistribution attacks
- 5 constraints on average to minimize the damage

After 5 iterations, the CRO-Exp algorithm identified the most important constraints for post-processing synthetic loads and **ensuring grid resilience**.

Post-attack damage as a function of constraints selected for post-processing optimization

- ▶ Synthetic grid data is optimized to guarantee privacy, quality and cyber resilience simultaneously
- ▶ Trade-offs under linear cost functions are “flat”: resilience is achieved with little to no impact on data quality
- ▶ The tri-level post-processing optimization can be efficiently collapsed to single-level optimization under reasonable and judicious approximations (connecting bi-level and robust optimization techniques)

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Synthesizing Grid Data With Cyber Resilience and Privacy Guarantees

Shengyang Wu[✉], Student Member, IEEE, and Vladimir Dvorkin[✉], Member, IEEE

Abstract—Differential privacy (DP) provides a principled approach to synthesizing data (e.g., loads) from real-world power systems while limiting the exposure of sensitive information. However, adversaries may exploit synthetic data to calibrate cyberattacks on the source grids. To control these risks, we propose new DP algorithms for synthesizing data that provide the source grids with both cyber resilience and privacy guarantees. The algorithms incorporate both normal operation and attack optimization models to balance the fidelity of synthesized data and cyber resilience. The resulting post-processing optimization is reformulated as a robust optimization problem, which is compatible with the exponential mechanism of DP to moderate its computational burden.

Index Terms—Power systems, synthetic dataset, differential privacy, cyber security.

with such releases remain largely unexplored. Possible cyber attacks include *false data injection*, which subtly alters state estimation results [13], *line outage masking*, which disconnects a transmission line and misguides a control center to seek outage elsewhere [14], and *load redistribution*, which manipulates demand measurements to increase OPF cost and constraint violation [15]. The latter is of main interest to this letter. Executing such attacks requires some grid knowledge [16], which is traditionally difficult to obtain. However, the availability of synthetic grid data may unintentionally inform adversaries and help them calibrate the attack.

Contribution: Recognizing the risks that synthetic grid parameters may inform cyber adversaries, we develop new DP algorithms that simultaneously guarantee cyber resilience and privacy for the source power grids. Our algorithms

Check paper for details on:

- ▶ DP guarantees of CRO and CRO-Exp
- ▶ Connection between Bi-level and RO
- ▶ Experiment settings, data and code